

Quan (Kyle) Zhang

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Education

Ph.D., Economics (Quantitative Economics & Econometrics), University of Hawai‘i at Mānoa 2023 – Present
Advisor: Sang-Hyop Lee Expected: 2027

M.S., Quantitative Health Sciences, John A. Burns School of Medicine, UH Mānoa 2024 – 2026

M.S., Industrial & Systems Engineering, Northern Illinois University 2012

Research Interests

Health economics and policy; causal inference and applied econometrics for health and social outcomes; **quantitative analytic methods for longitudinal cohort data with complex survey designs (weighting, stratification, replicate-weights variance estimation)**; population aging, frailty dynamics, and the joint determination of health and labor supply over the life course; disaster health economics and community resilience; quantitative health sciences and clinical research methodology, including cost-effectiveness analysis, psychometric measurement, and biomarker-integrated longitudinal study design; health equity and disparities.

Peer-Reviewed Publications

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1. Zhang, Q., & Fu, Y. (2026). Ensuring trustworthy AI assisted guideline development for clinical practice. *npj Digital Medicine*, 9(1), 661. [DOI](#)
 2. Zhang, Q. (2026). Do negative social ties accelerate aging in adults, or does aging erode social ties? *Proceedings of the National Academy of Sciences (PNAS)*, 123(20), e2608036123. [DOI](#)
 3. Zhang, Q. (2026). Keep REMS focused on safety, not as a barrier to generics. *Health Affairs*. [DOI](#)
 4. Zhang, Q. (2026). Enhancing life-course analyses of financial disadvantage and depressive mood through longitudinal design and causal inference. *Journal of Epidemiology*. [DOI](#)
 5. Zhang, Q. (2025). Overcoming bias in a fall prevention program in resource-limited areas. *JAMA*. [DOI](#)
 6. Zhang, Q. (2025). Operationalizing distributional cost-effectiveness in genomic medicine: A US-focused comment on Smith et al. *Value in Health*. [DOI](#)
 7. Zhang, Q. (2025). On “The joint impact of greenspace and air pollution on mortality”: Methodological proposals. *International Journal of Epidemiology*. [DOI](#)
 8. Zhang, Q. (2025). A causal roadmap for Child Opportunity Index 2.0. *Public Health Reports*. [DOI](#)

Papers Under Review

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1. Zhang, Q., & Hu, W. (2026). “Subclinical Cardiometabolic–Inflammatory Burden, Disability-Free Working Life and Work Exit in Three Ageing Cohorts.” Under review, *Nature Aging*. [SSRN](#)
 2. Zhang, Q. (2026). “Inflammatory, Nutritional, and Metabolic Biomarker Profiles in Early-Onset Colorectal Cancer: A 24-Year Cross-Sectional Analysis of NHANES, 2001–2023.” Under review, *JAMA Oncology*.
 3. Zhang, Q., & Lee, S.-H. (2026). “Adult Children’s Labor Supply and Co-residence after Parental Dementia: A Lifecycle Reassessment.” Under review, *Health Economics*.
 4. Zhang, Q. (2026). “Which Diagnoses Change How Long People Think They Will Live: Cancer against Other First Diagnoses in Three Ageing Panels.” Under review, *The Journal of the Economics of Ageing*. [SSRN](#)
 5. Zhang, Q. (2026). “Restricted-Scope Medi-Cal Enrollment Rose Sharply After California’s 2026 Adult Enrollment Freeze.” Under review, *Health Affairs Scholar*.
 6. “Network Toxicology and Multi-Omics Reveal Potential Mechanisms of DDT-Type Insecticides on Hepatocellular Carcinoma.” Under revision, *Discover Oncology*. (Co-author.)

GBD 2023 Collaborative Publications (Group Authorship, Under Review)

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1. GBD 2023 Colon and Rectum Cancer Collaborators (2026). “Global, Regional, and National Burden of Colon and Rectum Cancer from 1990 to 2023, with Forecasts to 2050: A Systematic Analysis for the Global Burden of Disease Study 2023.” Under review, *The Lancet Oncology*. (IHME ID: 4219; named collaborator-author.)
 2. GBD 2023 Western Pacific WHO AMR Collaborators (2026). “The Burden of Bacterial Antimicrobial Resistance in the WHO Region of the Western Pacific: A Cross-Country Systematic Analysis.” Under review, *The Lancet Regional Health – Western Pacific*. (IHME ID: 4006; named collaborator-author.)
 3. GBD 2023 Type 1 Diabetes Collaborators (2026). “Global, Regional, and National Type 1 Diabetes Burden: A Systematic Analysis of the Global Burden of Disease Study 2023.” Under review, *The Lancet Primary Care*. (IHME ID: 5338; named collaborator-author.)

Working Papers

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1. Zhang, Q. (2026). “Chronic versus Acute Health Shocks and the Margins of Labor Supply: Evidence from Frailty Dynamics in an Aging Population.”

2. Zhang, Q. (2026). “Health Shocks and the Permanent Labor-Market Exit of Older Europeans.”
3. “Sustained Insulin Resistance and Incident Coronary Heart Disease in Older English Adults: A Selection-Weighted Cohort Analysis of the English Longitudinal Study of Ageing.” [SSRN](#)
4. Zhang, Q. (2026). “Balancing Innovation Incentives and Equitable Access for AI Health Technologies: Policy Levers for IP, Transparency, and Solidarity.”
5. Zhang, Q. (2026). “How Much Care, and How Much Harm? Measuring Both After a Spouse Falls Ill.”
6. Zhang, Q. (2026). “Caregiving as Health Capital Depreciation: Time Allocation Evidence from IPUMS ATUS.”
7. Zhang, Q. (2026). “Falls, Missing Markets, and Labor Supply: Evidence from China.”
8. Zhang, Q. (2026). “Later, Longer, Fewer and the Quantity–Quality Trade-off: Evidence from CHARLS.”

Invited Contributions

1. Zhang, Q. (Forthcoming). “Multidimensional Poverty Index (MPI): Measurement, Health Deprivations, and Policy Use.” In R. Baikady (Ed.), *The Palgrave Encyclopedia of Global Poverty Research*. Palgrave Macmillan / Springer Nature.
2. Zhang, Q. (Forthcoming). “Social Determinants of Health and Multidimensional Poverty.” In R. Baikady (Ed.), *The Palgrave Encyclopedia of Global Poverty Research*. Palgrave Macmillan / Springer Nature.

Conference Presentations

1. Zhang, Q. (2026). “Chronic versus Acute Health Shocks and the Margins of Labor Supply: Evidence from Frailty Dynamics in an Aging Population.” Southern Economic Association (SEA) 96th Annual Meeting, Houston, TX, November 21–23, 2026.
2. Zhang, Q. (2026). “Nutritional Depletion, Not Inflammation Alone, Independently Associates with Colorectal Cancer: A 24-Year NHANES Analysis (2001–2023) with a Distinct Early-Onset Phenotype.” Flash talk, International Conference on Intelligent Biology and Medicine (ICIBM) 2026, Buffalo, NY, August 2026.

Research Experience

Graduate Research Assistant — UHERO, University of Hawai'i at Mānoa	2024 – Present
<i>Project 1: Maui Wildfire Exposure Study (MauiWES)</i>	2024 – Present
NIH- and state-funded longitudinal cohort. PI: Ruben Juarez. In collaboration with UHERO, JABSOM, and UH Maui College.	
- Analyzed mixed-methods data (biomarker, behavioral, socioeconomic) for 2,000+ participants in an IRB-compliant longitudinal study; generated individual-level risk forecasts informing disaster-recovery policy. - Designed quantitative instruments and contributed to participant recruitment protocols for ongoing cohort waves. Contributing to a study published in <i>JAMA Network Open</i> (2025).	
<i>Project 2: Roots Reborn Program Evaluation</i>	2025 – 2026
Post-wildfire economic empowerment intervention.	
- Designed and executed full evaluation of an IRB-approved human-subjects intervention: Qualtrics survey design → matched-panel construction → pre-post causal analysis → executive summary for policymakers. - Built session-level evaluation framework linking program costs to quality-adjusted participant outcomes, producing a submission-ready manuscript with actionable policy recommendations.	
<i>Project 3: Minority-Owned Small Business Financing Study</i>	2024 – 2025
City & County of Honolulu OER-funded.	
Built 4-module R causal-inference pipeline (PSM, IPW, Heckman selection, mediation), identifying heterogeneous treatment effects of financing barriers across business subgroups.	
GBD Collaborator — Global Burden of Disease Study 2023, IHME	2025 – Present
- Invited collaborator at the Institute for Health Metrics and Evaluation, University of Washington. Provided substantive peer review and contributions for nine GBD 2023 estimation manuscripts spanning cancer, cardiovascular and neurodegenerative disease, metabolic disease, dietary and environmental risks, antimicrobial resistance, and population ageing. - Contributions recognized with named collaborator-authorship on three GBD 2023 papers currently under review at <i>The Lancet Oncology</i>, <i>The Lancet Regional Health – Western Pacific</i>, and <i>The Lancet Primary Care</i>.	
Research Assistant (Co-Op) — Argonne National Laboratory, TRACC	2011
Developed regression models for urban evacuation timing in collaboration with NASA. Applied simulation, Design of Experiment, and statistical modeling for Chicago evacuation planning.	

Prior Industry Experience

Program Manager , Microsoft Corporation, SQL Server Data Warehousing (APS), Redmond, WA	2014 – 2015
Managed \$4M delivery pipeline of cloud data-warehouse appliances; developed logistics and cost model for Parallel Data Warehousing infrastructure.	
Project Management Specialist , Quanta Computer (Amazon AWS NPI), Bellevue, WA	2013 – 2014
Coordinated New Product Introduction releases for AWS cloud servers; managed BOM provisioning and cross-cultural supply chain operations between North American and Asian markets.	
Project Engineer (Co-Op) , Caterpillar Inc., Manufacturing Strategy, Aurora, IL	2012
Delivered NPV-based analysis projecting ~\$1M/year savings across warehouse-relocation scenarios; achieved 60% reduction in urgent-order tardiness via Six Sigma DMAIC.	

Teaching Experience

Instructor , Introduction to Statistics, UH Mānoa Department of Economics	Summer 2025
Teaching Assistant , Principles of Microeconomics & Macroeconomics, UH Mānoa	2023 – 2024
Teaching Assistant , Arena Simulation, Operations Research, Decision Analysis, NIU ISE	2010 – 2012

Awards & Honors

2026	Research Stipend, College of Social Sciences, University of Hawai'i at Mānoa
2026	AWARD Network Summer Institute, University of California, San Francisco (NIA/NIH-funded)
2026	Bernie Campbell Fellowship Fund Award, University of Hawai'i at Mānoa
2025	Burnham O. Campbell Fund Award, University of Hawai'i at Mānoa
2024, 2025	Hung Family Scholarship Award
2024	Jordan & Cara Odo Foundation Leadership Scholarship
2023	Achievement Award, Department of Economics, UH Mānoa

Professional Service

Peer Reviewer , <i>Public Health in Practice</i> (Elsevier), 2025–Present
Collaborator & Peer Reviewer , Global Burden of Disease Study 2023, Institute for Health Metrics and Evaluation (IHME), University of Washington, 2025–Present
Global Type 1 Diabetes (IHME ID: 5338), 2025 — <i>collaborator-author</i>

- Antimicrobial Resistance in the WHO Western Pacific Region (IHME ID: 4006), 2026 — *collaborator-author*
- Global Colon & Rectum Cancer (IHME ID: 4219), 2026 — *collaborator-author*
- Global Environmental Exposures (IHME ID: 6176), 2026
- Global Dietary Risks (IHME ID: 6115), 2026
- Global Ageing and Non-Communicable Diseases (IHME ID: 5410), 2026
- Global Alzheimer's Disease & Other Dementias (IHME ID: 6217), 2026
- Global Stroke (IHME ID: 6155), 2026
- Global Ageing and Inequality (IHME ID: 5409), 2026

Professional Memberships

American Economic Association (AEA) · American Society of Health Economists (ASHEcon) · International Health Economics Association (iHEA) · Society for Epidemiologic Research (SER) · Gerontological Society of America (GSA) · Hawaii Economic Association (HEA) · Pi Gamma Mu

Technical Skills

Statistical Software: R (tidyverse, fixest, modelsummary, survey, srvyr, ggplot2, Shiny), Stata 18 (including `svy` suite for complex-survey estimation), Python (pandas, statsmodels, scikit-learn), SQL, Git.

Complex Survey & Longitudinal Methods: Stratified multistage cluster designs, longitudinal and cross-sectional weighting, replicate-weights variance estimation (BRR, jackknife, Taylor linearization), wave-to-wave attrition adjustment, nonresponse weighting.

Causal Inference & Econometrics: Two-way fixed effects, difference-in-differences, instrumental variables (2SLS), event study designs, regression discontinuity, propensity score matching, inverse probability weighting, Heckman selection models, mediation analysis, sensitivity analysis (Oster 2019; Cinelli–Hazlett 2020), Lee bounds, specification curve analysis, wild cluster bootstrap, A/B testing frameworks, power analysis.

Machine Learning & Computational Methods: Double/debiased machine learning (DML), causal forests, LASSO/elastic net, multiple imputation (MICE).

Quantitative Health Sciences: Frailty index construction (Rockwood deficit accumulation), item response theory (IRT), confirmatory factor analysis (CFA), GAD-7/CES-D psychometric scoring, cost-effectiveness analysis (CEA/DCEA), survey design (Qualtrics).

Survey & Data Platforms: CHARLS, GBD, MEPS, HRS, BRFSS, NHANES, IPUMS (USA/CPS/International), ATUS, SHARE.

Certifications

Google Business Intelligence Professional Certificate · Google Data Analytics Professional Certificate · Certified Six Sigma DMAIC Green Belt (Caterpillar Inc.)

Languages

English (fluent) · Mandarin Chinese (native)